

A Neural Network Approach for Calibrating Memristor Crossbar Arrays

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Abstract—In-memory computing using memristor crossbar arrays is a promising approach to overcome data transfer bottlenecks in von Neumann architectures. However, practical deployment faces challenges from IR-drop, device variability, and parasitic effects, leading to accuracy degradation in analog matrix-vector multiplications. This work proposes a neural network-based calibration framework that compensates for such non-idealities post-fabrication, eliminating the need for hardware changes. The method learns systematic distortions and applies a lightweight, post-process calibration layer, achieving substantial accuracy improvements and robust performance under various signal conditions.

Index Terms—Memristor, Crossbar Array, Calibration, Neural Networks, In-Memory Computing, Analog Computing, IR-drop, Parasitic Effects.

I. INTRODUCTION

The traditional von Neumann architecture suffers from costly data movement, which impedes computational efficiency. In-memory computing with memristor crossbar arrays presents a compelling solution, as it enables computation directly where the data resides. Despite their promise, analog matrix-vector multiplication (MVM) implementations using memristor crossbars are impacted by issues such as IR-drop, parasitic capacitances and inductances, and intrinsic device variability. These factors result in significant output deviations from ideal computations, thus limiting the effectiveness of crossbar-based accelerators for practical applications.

Conventional solutions include hardware, circuit-level, and algorithmic approaches. Hardware fixes often increase area and design complexity, while circuit-level techniques may only offer partial corrections. Algorithmic solutions like retraining with hardware-aware models also present trade-offs and may not fully address non-idealities. Therefore, a robust, efficient, and hardware-agnostic solution is desirable.

II. MODULATOR DESIGN AND IMPLEMENTATION

The proposed solution employs a neural network-based calibration framework designed for post-fabrication compensation of systematic analog distortions. Key aspects include:

- **Input:** The framework takes in the ideal conductance matrix, encodes it into a latent vector, and combines it with real (measured) output currents from the memristor crossbar.
- **Neural Calibration:** Through calibration layers with nonlinear activations (e.g., sigmoid), the network learns a non-linear inverse mapping that compensates for IR-drop, parasitic, and nonlinearity effects.
- **Training:** The system is trained offline using a large dataset composed of random conductance matrices (1,000 matrices ranging from 16.6 μ S to 100 μ S), each with 100 random input vectors (0.1–0.5V), resulting in 100,000 VMM samples.

- **Implementation:** Data is collected from a custom Verilog-A memristor model integrated within Cadence Virtuoso using 22nm FD-SOI CMOS.
- **Flexibility:** The approach is adaptable to device drift and aging with a small calibration set without frequent retraining.

III. EXPERIMENTAL RESULTS AND SYSTEM DEMONSTRATION

Experimental validation demonstrates that the neural network calibration yields about $1000\times$ reduction in mean squared error (MSE): from 0.13 before calibration to 1.4×10^{-4} after calibration. The method generalizes well to unseen signals, handling both sinusoidal and DC inputs.

The framework effectively compensates for systematic non-idealities and demonstrates robust performance even in the presence of noise. Its lightweight nature ensures low overhead, as all inference occurs at runtime after offline training. Compared to device-level and circuit-level solutions, this post-fabrication approach offers scalability and does not incur additional hardware complexity or significant area cost.

IV. CONCLUSION

This work introduces a neural network-based framework that restores the accuracy of memristor crossbar arrays by correcting analog distortions without requiring hardware changes. The demonstrated strategy greatly improves the feasibility of analog in-memory computing and supports the development of high-precision neuromorphic systems. Future efforts include silicon validation under varying noise and temperature conditions, extension to large-scale or tiled arrays, and public release of the calibration core for community use.